Exp. No. 26

Write a LEX program to count the number of comment lines in a given C program and eliminate them and write into another file.

Input Source File: (input.c)

```
#include<stdio.h>
int main()
{
int a,b,c; /*varible declaration*/
printf("enter two numbers");
scanf("%d %d",&a,&b);
c=a+b;//adding two numbers
printf("sum is %d",c);
return 0;
}
```

Program: (comment.l)

```
%{
int com=0;
%}
%s COMMENT
%%
"/*" {BEGIN COMMENT;}
<COMMENT>"*/" {BEGIN 0; com++;}
<COMMENT>\n {com++;}
<COMMENT>. {;}
\/\/.* {; com++;}
.|\n {fprintf(yyout,"%s",yytext);}
%%
void main(int argc, char *argv[])
{
if(argc!=3)
{
printf("usage : a.exe input.c output.c\n");
exit(0);
}
yyin=fopen(argv[1],"r");
yyout=fopen(argv[2],"w");
yylex();
printf("\n number of comments are = %d\n",com);
}
int yywrap()
```

```
{
return 1;
}
```

Output:

```
C:\Users\Administrator\Documents\exp 26.exe

G:\lex>flex comment.l
G:\lex>gcc lex.yy.c
G:\lex>a.exe input.c output.c
number of comments are = 2

G:\lex>type output.c
#include<stdio.h>
int main()
{
int a,b,c;
printf("enter two numbers");
scanf("%d %d",&a,&b);
c=a+b;
printf("sum is %d",c);
return 0;
}

G:\lex>_
```

Exp. No. 27

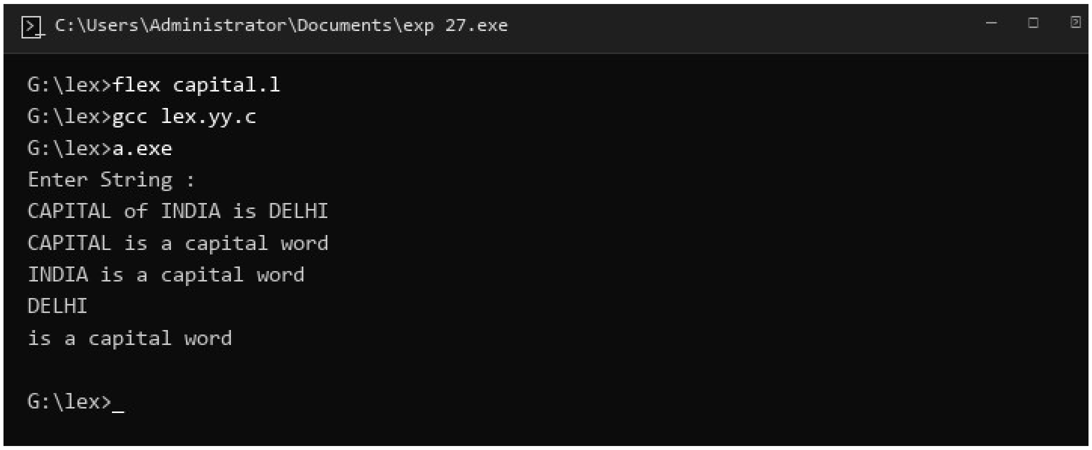
Write a LEX program to identify the capital words from the given input.

Program: (capital.l)

```
%%
[A-Z]+[\t\n ] { printf("%s is a capital word\n",yytext); }
. ;
%%

int main( )
{
        printf("Enter String :\n");
        yylex();
}
int yywrap( )
{
        return 1;
}
```

Output:



```
C:\Users\Administrator\Documents\exp 27.exe
G:\lex>flex capital.l
G:\lex>gcc lex.yy.c
G:\lex>a.exe
Enter String :
CAPITAL of INDIA is DELHI
CAPITAL is a capital word
INDIA is a capital word
DELHI
is a capital word

G:\lex>_
```

Exp. No. 28

Write a LEX Program to check the email address is valid or not.

Program: (email_valid.l)

```
%{
int flag=0;
%}
%%
[a-z . 0-9]+@[a-z]+".com"|".in" { flag=1; }
%%
int main()
{
yylex();
if(flag==1)
printf("Accepted");
else
printf("Not Accepted");
}
int yywrap()
{ return 1;
 }
```

Output:

```
C:\Users\Administrator\Documents\exp 28.exe

G:\lex>flex email_valid.l
G:\lex>gcc lex.yy.c
G:\lex>a.exe
sse123@gmail.com
Accepted

G:\lex>_
```

Exp. No. 29

Write a LEX Program to convert the substring abc to ABC from the given input string

Program: (substring.l)

```
%{
int i;
%}
%%
[a-z A-Z]* { for(i=0;i<=yyleng;i++)
          { if((yytext[i]=='a')&&(yytext[i+1]=='b')&&(yytext[i+2]=='c'))
            { yytext[i]='A';
             yytext[i+1]='B';
             yytext[i+2]='C';
            }
          }
         printf("%s",yytext);
        }

[\t]* return 1;
.* {ECHO;}
\n {printf("%s",yytext);}
%%
int main()
{
yylex();
}
int yywrap()
{
return 1;
}
```

Output:

```
C:\Users\Administrator\Documents\exp 29.exe

G:\lex>flex substring.l
G:\lex>gcc lex.yy.c
G:\lex>a.exe
abcdefghabcijkla
ABCdefghABCijkla

G:\lex>_
```

Exp. No. 30

Implement a LEX program to check whether the mobile number is valid or not.

Program: (mobile.l)

```
%%
[1-9][0-9]{9}  {printf("\nMobile Number Valid\n");}
.+ {printf("\nMobile Number Invalid\n");}
%%
int main()
{
	printf("\nEnter Mobile Number : ");
	yylex();
	printf("\n");
	return 0;
}
int yywrap()
{ }
```

Output:

```
C:\Users\Administrator\Documents\exp 30.exe

G:\lex>flex mobile.l
G:\lex>gcc lex.yy.c
G:\lex>a.exe
Enter Mobile Number : 7856453489
Mobile Number Valid

G:\lex>_
```